An Exact Rooted Sum-of-Squares Certificate for the House Graph

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Abstract

We prove the chromatic-polynomial lower bound for the house graph, NetworkX Atlas 43, on its entire required interval. The proof is a direct graphon sum-of-squares identity with two rational positive-semidefinite Gram matrices. Its machine-readable certificate is checked coefficient by coefficient in exact rational arithmetic. It uses ordinary non-induced homomorphism densities and no flag-algebra rules.

1 Statement

Let H be the house graph and let W be an arbitrary graphon of edge density $p = t(K_2, W)$. Its chromatic target is

$$\Phi_H(p) = p(2p-1)(3p^2-3p+1) = 6p^4 - 9p^3 + 5p^2 - p.$$

Theorem 1. *For every graphon W and every $p \in [1/2, 1]$,*

$$t(H, W) \geq p(2p-1)(3p^2-3p+1).$$

Thus Atlas 43 has positive classification.

2 The rooted basis and the exact identity

Use three labelled vertices $0, 1, 2$ and one unlabelled branch vertex 3. There are six possible edges on these four vertices. For each subset S of those edges let F_S be the corresponding partially labelled simple graph. After fixing the labelled variables $x = (x_0, x_1, x_2)$, sample one shared Bernoulli bit $\xi_{ij} \in \{0, 1\}$ for each of the three labelled edges, independently conditional on x, with

$$\mathbb{P}(\xi_{ij} = 1 \mid x) = W(x_i, x_j).$$

Write

$$f_S(x, \xi) = \prod_{\substack{ij \in S \\ i,j<3}} \xi_{ij} \int_\Omega \prod_{\substack{i3 \in S \\ i<3}} W(x_i, x_3)\, \mathrm{d}x_3,$$

where x_3 is integrated. Order the $2^6 = 64$ subsets by their binary edge masks and put $f(x, \xi) = (f_S(x, \xi))_S$.

Set

$$u = 2p - 1 \in [0, 1], \qquad \widehat{f}(x, \xi) = \begin{pmatrix} f(x, \xi) \\ u f(x, \xi) \end{pmatrix}.$$

The certificate, embedded in the Lean modules of `lean/Taeyoung/Methods/RootedSOS/`, consists of integer matrices

$$F_0 \in \mathbb{Z}^{128 \times 107}, \qquad F_1 \in \mathbb{Z}^{64 \times 48},$$

a common positive integer $D = 10^6$, and exact rational symmetric matrices

$$Z_0 \in \mathbb{Q}^{107\times 107}, \qquad Z_1 \in \mathbb{Q}^{48\times 48}.$$

The matrices Z_i are stored sparsely as $I + C_i$. Define

$$Q_0 = D^{-2} F_0 Z_0 F_0^{\mathsf{T}}, \qquad Q_1 = D^{-2} F_1 Z_1 F_1^{\mathsf{T}}.$$

The exact certificate asserts the identity

$$\boxed{t(H,W) - \Phi_H(p) = \int_{\Omega^3} \mathbb{E}_{\xi|x}\left[\widehat{f}(x,\xi)^{\mathsf{T}} Q_0 \widehat{f}(x,\xi)\right]\,\mathrm{d}x + u(1-u)\int_{\Omega^3} \mathbb{E}_{\xi|x}\left[f(x,\xi)^{\mathsf{T}} Q_1 f(x,\xi)\right]\,\mathrm{d}x.} \tag{1}$$

This is an ordinary graphon identity. Indeed, $\int \mathbb{E}_{\xi|x}[f_S(x,\xi) f_T(x,\xi)]\,\mathrm{d}x$ is the homomorphism density of the simple graph obtained by taking two copies of F_S, F_T and identifying equally labelled vertices. A labelled edge in both flags is counted once because $\xi_{ij}^2 = \xi_{ij}$, and conditional averaging then replaces every surviving ξ_{ij} by $W(x_i, x_j)$. This shared Bernoulli step is essential: directly using graphon values on labelled edges would incorrectly replace $W(x_i, x_j)^2$ by $W(x_i, x_j)$ when the two flags share that edge. If the glued graph has m isolated-edge components, the exact fixed-density identity

$$t(G \sqcup mK_2, W) = p^m t(G, W)$$

removes them. Expanding $p = (1+u)/2$ leaves 91 nonzero coefficients, and the certificate checker verifies every one of them over $\mathbb{Q}$.

3 Positive semidefiniteness and exact verification

For a symmetric matrix $Z = (z_{ij})$, strict diagonal dominance with positive diagonal,

$$z_{ii} - \sum_{j\neq i} |z_{ij}| > 0 \quad \text{for every } i,$$

implies positive definiteness. Exact fraction arithmetic gives the minimum row margins

$$\min_i \left((Z_0)_{ii} - \sum_{j\neq i} |(Z_0)_{ij}|\right) > 0.9997, \qquad \min_i \left((Z_1)_{ii} - \sum_{j\neq i} |(Z_1)_{ij}|\right) = \frac{999955414399}{10^{12}} > 0.$$

Consequently Z_0, Z_1 are positive definite and Q_0, Q_1 are positive semidefinite.

The certificate was checked independently in exact rational arithmetic; the checks are the following:

1. NetworkX Atlas 43 has target $6p^4 - 9p^3 + 5p^2 - p$ by its chromatic polynomial;
2. the dimensions and sparse entries of the two Gram factors are valid;
3. both diagonal-dominance margins are positive rational numbers; and
4. all 91 coefficients in (1) agree exactly after isolated edge components are replaced by powers of p.

Running it prints

```
verified Atlas 43 exact interval SOS: equations=91 orders=[107, 48]
```

and exits successfully. No floating-point value is used by the checker.

Proof of Theorem 1. Both quadratic expressions inside the conditional expectations on the right of (1) are nonnegative, and $u(1-u) \geq 0$ for $p \in [1/2, 1]$. Hence the right-hand side is nonnegative, proving the theorem. □

4 How the certificate was recovered

The initial numerical SDP is singular because every balanced complete k-partite graphon is an equality case at $p = 1 - 1/k$. Rooted evaluations on those Turan graphons give exact rational kernel spaces of dimensions 15 and 10. Two further six-dimensional faces have half-integral bases. After restricting to the exact complements of these spaces, the reduced Gram orders are 107 and 48 and the numerical feasibility margin is $0.0056368\ldots$. A rational Cholesky preconditioner then converts the final exact correction into the strictly diagonally dominant matrices above.

The discovery tools are not assumptions of the proof: the certificate contains all data needed to verify (1) and positivity.

Remark 2. *This construction is a direct rooted quantum-graph sum of squares. It does not use induced flags, asymptotic flag-density multiplication, or a flag-algebra semidefinite certificate. Every product in* (1) *is interpreted directly as an integral of graphon kernels.*